

Taller 1, punto 6: perceptron sobre las particiones de banknote

```
clear; close all; clc
load banknote_splits.mat
```

Parametros

```
regla = 3; umbral = 0; salida = [-1 1]; max_epocas = 200; semilla = 1;
alphas = [0.01 0.05 0.1 0.5 1];
```

Entrenamiento con y sin escalar las entradas

```
R = table();
for escalar = [false true]
    for k = 1:4
        Xtr = S(k).Xtr; Xte = S(k).Xte;
        dtr = 2*S(k).ytr - 1; dte = 2*S(k).yte - 1;
        if escalar
            mn = min(Xtr); mx = max(Xtr);
            Xtr = (Xtr - mn) ./ (mx - mn); Xte = (Xte - mn) ./ (mx - mn);
        end
        for a = alphas
            [w, b, ep, conv] = perceptron(Xtr, dtr, regla, a, umbral, salida,
max_epocas, semilla);
            acc_tr = 100*mean(predecir(Xtr, w, b, umbral, salida) == dtr);
            acc_te = 100*mean(predecir(Xte, w, b, umbral, salida) == dte);
            R = [R; table(escalar, k, {sprintf('%d-%d', 100*S(k).p,
100-100*S(k).p)}, a, ep, conv, acc_tr, acc_te, ...
                'VariableNames',
{'Escalado','k','Particion','Alpha','Epocas','Convergio','Exact_train','Exact_test'})];
        end
    end
end
R
```

R = 40x8 table

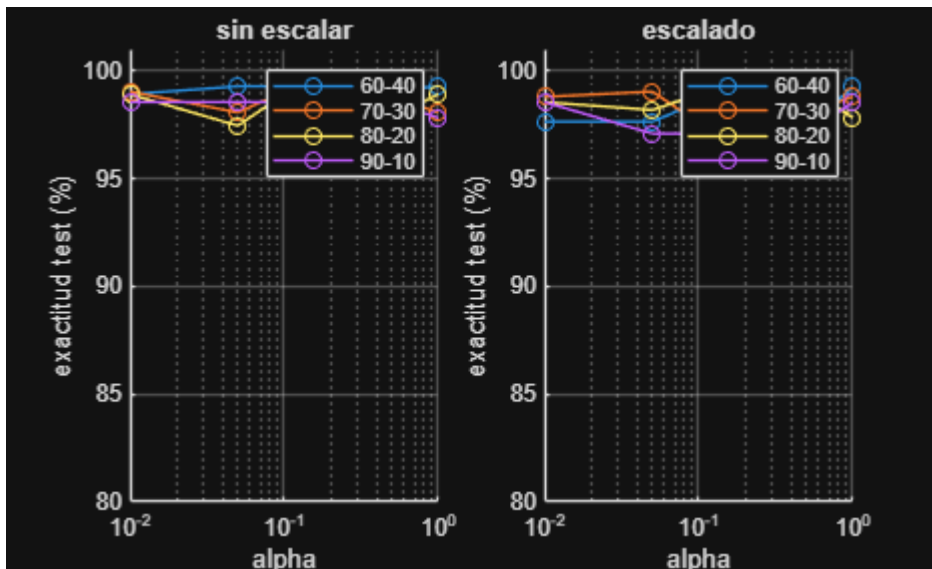
	Escalado	k	Particion	Alpha	Epocas	Convergio	Exact_train
1	false	1	'60-40'	0.0100	200	false	98.6634
2	false	1	'60-40'	0.0500	200	false	98.9064
3	false	1	'60-40'	0.1000	200	false	98.9064
4	false	1	'60-40'	0.5000	200	false	99.0279
5	false	1	'60-40'	1	200	false	98.7849
6	false	2	'70-30'	0.0100	200	false	98.9583

	Escalado	k	Particion	Alpha	Epocas	Convergio	Exact_train
7	false	2	'70-30'	0.0500	200	false	98.0208
8	false	2	'70-30'	0.1000	200	false	99.1667
9	false	2	'70-30'	0.5000	200	false	99.0625
10	false	2	'70-30'	1	200	false	97.7083
11	false	3	'80-20'	0.0100	200	false	98.7250
12	false	3	'80-20'	0.0500	200	false	98.7250
13	false	3	'80-20'	0.1000	200	false	98.6339
14	false	3	'80-20'	0.5000	200	false	98.8160

⋮

Exactitud en test contra alpha

```
figure
titulos = {'sin escalar', 'escalado'};
for e = 1:2
    subplot(1,2,e); hold on
    for k = 1:4
        m = R.Escalado == (e==2) & R.k == k;
        plot(R.Alpha(m), R.Exact_test(m), '-o', 'DisplayName',
R.Particion{find(m,1)});
    end
    set(gca, 'XScale', 'log'); grid on; ylim([80 101]); legend
    xlabel('alpha'); ylabel('exactitud test (%)'); title(titulos{e});
end
```



Matriz de confusion del mejor caso

```
[~, i] = max(R.Exact_test);
R(i, :)
```

```
ans = 1x8 table
```

	Escalado	k	Particion	Alpha	Epoas	Convergio	Exact_train
1	false	1	'60-40'	0.0500	200	false	98.9064

```
k = R.k(i);
Xtr = S(k).Xtr; Xte = S(k).Xte; dtr = 2*S(k).ytr - 1; dte = 2*S(k).yte - 1;
if R.Escalado(i)
    mn = min(Xtr); mx = max(Xtr);
    Xtr = (Xtr - mn) ./ (mx - mn); Xte = (Xte - mn) ./ (mx - mn);
end
[w, b] = perceptron(Xtr, dtr, regla, R.Alpha(i), umbral, salida, max_epocas,
semilla);
figure
confusionchart(dte, predecir(Xte, w, b, umbral, salida))
```

